

Wrangling with dplyr library

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Data wrangling with dplyr library

The Wrangling

dplyr library first work on the first line then to the others line.

```
# First import the dataset
library(readr)
fertilizer_yield_data <- read_csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/1. Data Wrangling with dplyr library/fertilizer_yield_data.csv")
```

Rows: 20 Columns: 2

-- Column specification -----

Delimiter: ","

chr (1): fertilizer

dbl (1): yield

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.4.3

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
overall_mean <- mean(fertilizer_yield_data$yield, na.rm=TRUE)

summary_table <- fertilizer_yield_data %>%
  filter(yield>2) %>%
  group_by(fertilizer) %>% #Group the column based on the distinct value
  summarise(ava_yield = mean(yield, na.rm=TRUE)) %>% #Adding a new column and remove all the
  mutate(yield_lavel = ifelse(ava_yield>overall_mean,"High","Low")) #Add a new column from t

#Now see what we did
print(summary_table)
```

```
# A tibble: 4 x 3
  fertilizer ava_yield yield_lavel
  <chr>      <dbl> <chr>
1 Compost    2.33 Low
2 DAP        3.5  High
3 NPK        4.03 High
4 Urea       3.06 Low
```

Power of summarise

Summarise function can add multiple condition and it will give the new column every time.

```
New_table <- fertilizer_yield_data %>%
  group_by(fertilizer) %>%
  summarise(
    avg_yield = mean(yield, na.rm=TRUE),
    max_yield = max(yield, na.rm = TRUE),
    yield_sd = sd(yield, na.rm = TRUE),
    unique_yield = n_distinct(yield, na.rm=TRUE)
  )

print(New_table)
```

```
# A tibble: 4 x 5
  fertilizer avg_yield max_yield yield_sd unique_yield
  <chr>      <dbl>    <dbl>    <dbl>      <int>
1 Compost    2.33      2.6     0.222        4
2 DAP        3.5       3.9     0.483        4
3 NPK        4.03      4.2     0.171        4
4 Urea       3.06      3.4     0.351        5
```

Save the file as csv format

```
write.csv(summary_table, "Summary table.csv", row.names = FALSE)
```

Remove the unwanted column form the dataset

```
fertilizer_yield_data %>%
  select(fertilizer, yield) #Just Show the written column
```

```
# A tibble: 20 x 2
  fertilizer yield
  <chr>      <dbl>
1 Urea      2.5
2 Urea      3.1
3 DAP       2.8
4 DAP       3.6
5 NPK       4
6 NPK       3.8
```

7	Compost	2.2
8	Compost	NA
9	Urea	3.4
10	DAP	3.9
11	NPK	4.2
12	Compost	2.6
13	Urea	3
14	DAP	NA
15	NPK	4.1
16	Compost	2.4
17	Urea	3.3
18	DAP	3.7
19	NPK	NA
20	Compost	2.1

Rename the dataset column

```
fertilizer_yield_data %>%
  rename(fert= fertilizer)
```

```
# A tibble: 20 x 2
   fert    yield
  <chr>  <dbl>
1 Urea    2.5
2 Urea    3.1
3 DAP     2.8
4 DAP     3.6
5 NPK      4
6 NPK     3.8
7 Compost 2.2
8 Compost NA
9 Urea    3.4
10 DAP    3.9
11 NPK    4.2
12 Compost 2.6
13 Urea    3
14 DAP    NA
15 NPK    4.1
16 Compost 2.4
17 Urea    3.3
```

```

18 DAP      3.7
19 NPK      NA
20 Compost  2.1

```

Case_when

```

fertilizer_yield_data %>%
  mutate(yield2=yield*5) %>%
  mutate(yeild_level=case_when(
    yield2 < 10 ~ "Low",
    yield2 < 15 ~ "modarate",
    yield2 < 7 ~ "high",
    TRUE ~ "durjoy"
  ))

```

```

# A tibble: 20 x 4
  fertilizer yield yield2 yeild_level
  <chr>      <dbl> <dbl> <chr>
1 Urea      2.5   12.5 modarate
2 Urea      3.1   15.5 durjoy
3 DAP       2.8   14   modarate
4 DAP       3.6   18   durjoy
5 NPK       4     20   durjoy
6 NPK       3.8   19   durjoy
7 Compost   2.2   11   modarate
8 Compost   NA     NA   durjoy
9 Urea      3.4   17   durjoy
10 DAP      3.9   19.5 durjoy
11 NPK      4.2   21   durjoy
12 Compost  2.6   13   modarate
13 Urea     3     15   durjoy
14 DAP      NA     NA   durjoy
15 NPK      4.1   20.5 durjoy
16 Compost  2.4   12   modarate
17 Urea     3.3   16.5 durjoy
18 DAP      3.7   18.5 durjoy
19 NPK      NA     NA   durjoy
20 Compost  2.1   10.5 modarate

```